

Project: ISO JTC1/SC22/WG21: Programming Language C++
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Issues:
Previous: P3828R0

Rename the to_input view to as_input

The current name of the "to_input" view is misleading, because this sounds like the elements of the underlying range are actively processed (like with `std::ranges::to`). Instead it only ensures that the underlying range is seen **as** something different for those that later process the elements using this view. For views like this we usually use the name "as_..." (e.g. `as_const` or `as_rvalue`).

This paper proposes to rename the to_input view to as_input. This is both more intuitive and consistent.

Note that in this case this might include to also change the name of the section in the C++ standard and the name of the feature test macro.

Proposed Wording

(All against N4944)

In principle simply rename all occurrences of "to_input" to "as_input" and "to input view" to "as input view."

In **17.3.2 Header <version> synopsis [version.syn]**

Rename as follows:

```
#define __cpp_lib_ranges_toas_input 202502L // freestanding, also in <ranges>
```

In **25.2 Header <ranges> synopsis [ranges.syn]**

Rename as follows:

```
// 25.7.35, to as input view
template<input_range V>
    requires view<V>
class toas_input_view;
template<class V>
    constexpr bool enable_borrowed_range<toas_input_view<V>> =
        enable_borrowed_range<V>;
namespace views { inline constexpr unspecified toas_input = unspecified ; }
```

Rename

25.7.35 ~~To~~ input view [range.~~to~~.input]

to

25.7.35 **As input view [range.as.input]**

Rename

25.7.35.1 Overview [range.~~to~~.input.overview]

to

25.7.35.1 Overview [range.as.input.overview]

and rename inside:

- 1 ~~to~~as_input_view presents a view of an underlying sequence as an input-only non-common range.
[Note 1 : This is useful to avoid overhead that can be necessary to provide support for the operations needed for greater iterator strength. —end note]
- 2 The name `views::toas_input` denotes a range adaptor object (25.7.2). Let `E` be an expression and let `T` be `decltype((E))`. The expression `views::toas_input(E)` is expression-equivalent to:
 - (2.1) — `views::all(E)` if `T` models `input_range`, does not satisfy `common_range`, and does not satisfy `forward_range`.
 - (2.2) — Otherwise, `toas_input_view(E)`.

Rename

25.7.35.2 Class template ~~to~~as_input_view [range.~~to~~.input.view]

to

25.7.35.2 Class template as_input_view [range.as.input.view]

and rename inside:

```
namespace std::ranges {
    template<input_range V>
        requires view<V>
    class toas_input_view : public view_interface<toas_input_view<V>> {
        V base_ = V(); // exposition only
        // 25.7.35.3, class template toas_input_view::iterator
        template<bool Const> class iterator ; // exposition only
    public:
        toas_input_view() requires default_initializable<V> = default;
        constexpr explicit toas_input_view(V base);

        ...

        template<class R>
            toas_input_view(R&&) -> toas_input_view<views::all_t<R>>;
    }
    constexpr explicit toas_input_view(V base);
    1 Effects: Initializes base_ with std::move(base).
```

Rename

25.7.35.3 Class template ~~to~~as_input_view::iterator [range.~~to~~.input.iterator]

to

25.7.35.3 Class template as_input_view::iterator [range.as.input.iterator]

and rename inside:

```
namespace std::ranges {
    template<input_range V>
        requires view<V>
    template<bool Const>
    class toas_input_view<V>::iterator {
        using Base = maybe_const<Const, V>; // exposition only

        ...
    }
```

Acknowledgements

Thanks to a lot to everybody who helped and gave support to come to finally get this proposal done.

Rev0:

First initial version.

Rev1:

Fixed two typos according to LWG review.